Explaining without predicting

Anomaly detection and precedent retrieval for verifiable financial alerts

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Someone opens the app, sees this, and is left with three questions

NVDA

-4.1%

1. **Is this unusual?**
   4% is a lot for a quiet company. For another, it is a Tuesday.
2. **Was it the company, or the market?**
   If everything fell, the story is a different one.
3. **Has it happened before, and what followed?**
   Were there similar days? What did the price do afterwards?

Current free applications answer none of them.

Recent products claim to go further. None shows the past cases, one by one, with the measured outcome.

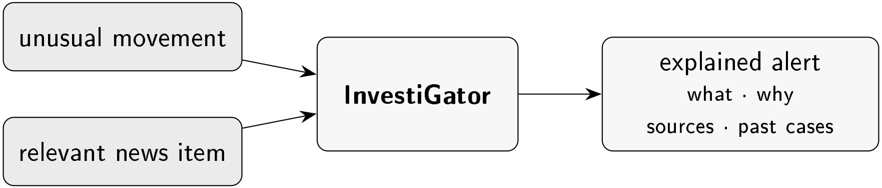
What exists today

	unusual?	company or market?	happened before?	cost
Broker application	—	—	—	included
News aggregator	—	—	—	free
Automated portfolio manager	—	—	—	% of portfolio
Professional terminal	✓	✓	✓	not published
Generated summary (2025–26)	*claim to answer; do not show the evidence*			free
This work	✓	✓	✓	**free**

Two recent products claim to answer the same question. Neither claims to **show the evidence**: the past cases, one by one, with date and measured outcome.

The difference I claim is not one of fluency: it is one of verifiability.

What the system does



And there is one thing it **refuses** to do.

It does not forecast prices

It never says the price will rise.

It never says to buy.

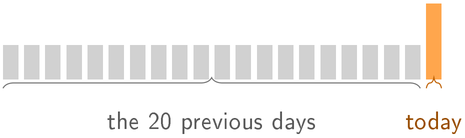
It never shows a price target.

It looks like a limitation. It is the opposite:
it is what makes everything else verifiable.

A statement about what already happened is checked against the record.
A forecast cannot be checked when it is read.

Technique 1: is it unusual?

Compare today with **the 20 previous days of the same stock**.

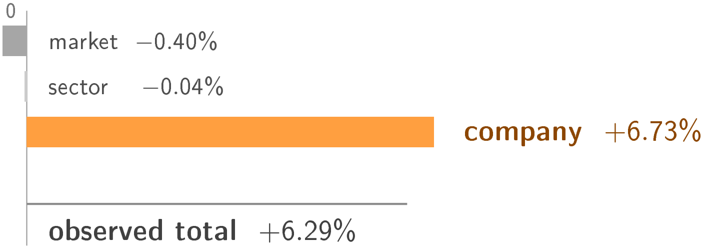


$$z = \frac{\text{today's move} - \text{mean}}{\text{standard deviation}}$$

$z = +7$ means the same thing for a quiet company and for a volatile one.
The day being judged is excluded from the norm it is judged against. Without that, it would be cheating.

Technique 2: was it the company, or the market?

A +6.29% can be the company or it can be the market. The broker application shows **the same number** in both cases.



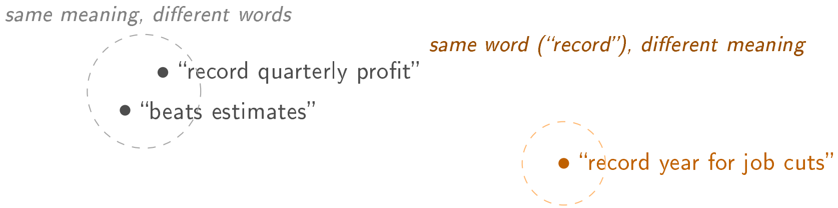
AMD rose **despite** the market and the sector pulling the other way. The *betas* are estimated **from previous days only**.

Caveat: the parts always add up, because the company's is the residual, and that does not prove they are well estimated. Median R^2 of 0.460, and negative in one case.

It is the only one of the four without a research question: there is no correct answer to compare against, so what is measured is the fit, not the accuracy.

Technique 3: has it happened before?

Searching by words fails in **both** directions:



Each headline becomes **384 numbers**: a coordinate of meaning.
They are compared by the **angle** between them.

Then, for each past case, what the price did at +1, +3 and +5 days is measured.
It is a measurement of the past, not a forecast.

Technique 4: is it worth interrupting?

Here there is no obvious formula. This is the place to **learn from the data**.

- **79 753** examples built from news and prices (2018–2023)
- The question: *after this headline, was there an abnormally large move?* **In absolute value**, never in which direction

Split by **whole days** with a **five-day** embargo, never by rows: two headlines from the same day cannot fall on opposite sides. And the guarantee is not promised, it is **enforced by a test**: two absurd futures (+300% and −99%) over the same past must produce **identical** inputs and **different** labels. If the future leaked, the inputs would change.

One real alert, from start to finish

Meta, 12 July 2026. *"Mark Zuckerberg Said Meta's AI Bets 'Haven't Come to Fruition Yet' as Shares Fell 5%"*

	stage	**what happened**
2	names the company?	"Meta", "Zuckerberg" ✓
3	is it recent?	less than 2 days ✓
4	similar cases	MSFT 0.46 · NVDA 0.51 · NVDA 0.46
5	evidence strong enough?	best $0.51 \geq 0.45$ ✓
6	worth interrupting?	$54\% \geq 50\%$ ✓
9	sent	to the channel, and stored

The outcomes went **in opposite directions**: $+2.70\% \cdot -3.87\% \cdot +5.05\%$

That is why they are shown **one by one**: an average would hide it.

Result 1: detection **yes**

A good detector should fire at a **similar** rate across different companies.

	min	**max**	**spread**
z-score	0.016	0.031	0.015
Fixed threshold of 3%	0.009	0.353	0.344

More than **20 times** more consistent.

And it also beats two learned methods: F_1 of 0.530 against 0.269 and 0.280.

Result 2: precedents yes

method	precision@5
By meaning (the method)	0.514
Always the largest sector	0.467
By shared words	0.346
At random (the floor)	0.240
The most recent	0.126

- Half the corpus is technology: **always returning technology** is worth 0.467 with no model at all. The real margin is +0.047, not +0.274.
- The method beats chance in all **five sectors**: 0.448 in energy, with a floor of 0.072. No advantage over a **filter using the query's known sector** was measured.
- Requiring earlier candidates: **0.513**, floor **0.259**, margin +0.254. The test does not impose the eight-day maturation used in production.
- **Similarity of theme does not guarantee direction**. Cases are shown one by one.

Result 3: triage **no**

model	PR-AUC
Volatility only	0.542
Context only	0.538
Context + text	0.496
Text only	0.439

No model that reads the text beats volatility.

It was the opposite of what I expected. I checked it three ways: a re-test under conditions favourable to the text (0.533, still below), resampling by groups, and **nine** different definitions of the label: volatility wins in all nine.

The mistake I found in my own work

I had written that triage *"almost quadruples"* precision: from 0.163 to 0.632.

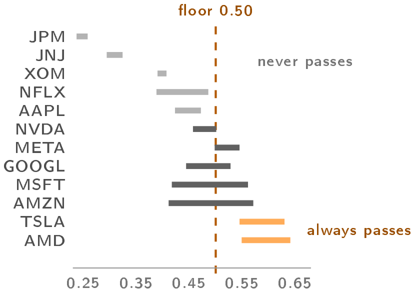
The 0.163 came from the "always alert" model, which gives everything the same score. **With everything tied, the metric breaks ties by file order**, and the file is sorted by date and by company name.

That number was not measuring choosing blindly.
It was measuring choosing alphabetically.

Real chance gives 0.379 $\Rightarrow$ the gain is **1.67**$\times$, not almost four.
And a table of **13 constants** gives 0.662: it beats the trained model.

And I found it again, one level up

The system decided whether to alert with a **fixed threshold** on the model's score: anything above 0.50 passes.



In **48%** of the distinct headlines, the outcome was determined by the **company** before the headline was read.

In no logged assessment did Apple reach the floor, with a maximum of 0.472; AMD exceeded it in all of them. Counted per decision the fraction rises to 60%, inflated by the minute-by-minute re-scoring.

And the obvious fix was wrong too

The temptation: if a fixed threshold measures the company and not the news item, make it *relative to each company*, which is exactly what the z-score did for prices.

I simulated it **before** writing any code. Half of it is solved: all twelve become represented.

But in companies whose score is nearly constant the percentile falls *on top of* the constant and almost everything ties above it. JNJ would pass **874** times.

A defect is not fixed by introducing the neighbouring one.

What was done: the model stopped deciding and started **ranking**, under a budget of 5 alerts per day. And that closed a divergence: the dissertation *evaluated* a budget policy while the system *deployed* a threshold one.

So what is the model worth? Two measurements that answer

1. I took the news away.

A *lookup table*: one fixed number per company, zero information about the headline.

what it sees	PR-AUC
the deployed one, nine inputs	0.538
the company only	0.534
nothing about the company	0.378
headline length only	0.378

A difference of 0.004, and 0.378 is the floor. **The deployed model is a lookup table.**

2. I compared the whole system.

Precision over the five news items the product shows per day.

policy	prec.@5
at random	0.375
whoever moved most today	0.489
the system	0.632
volatility, thirteen constants	0.662
oracle	0.968

It beats the realistic alternative. **But the oracle is the new information:** the important news *is* in the batch.

The missing margin is not in better data. It is in **separating two news items from the same company on the same day**.

“And where is the artificial intelligence engineering?”

I start with the uncomfortable part: this work trains **one** model, and it is a logistic regression with nine inputs that **lost** to a single-variable baseline.

The engineering is not in the size of the model.
It is in what had to be built so that that number, and the negative conclusion it supports, can be *trusted*.

1. The dataset is **built**, not found: 79 753 rows that did not exist.

2. The label is a **decision**, which is why **nine** alternative definitions were measured.

3. The temporal separation is enforced by a **test** that mutates the future, not promised.

4. The probabilities are calibrated, and their bias is **measured**.

5. The model, the calibrator and the run are **versioned**.

6. The deployed model **is** the evaluated model: exported to a lightweight format, with the fidelity of the conversion measured rather than assumed.

7. What runs is **observed**: every decision is logged and confronted days later with what the price did. That is how it became known that it was *not* helping.

The hard part was not choosing the technique. It was building the evaluation that **can say no**, and then believing it.

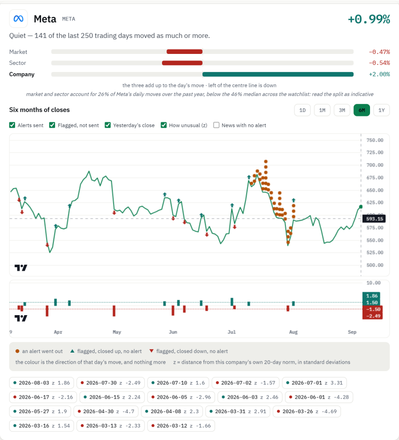
Limitations, said before being asked, and what closes them

Every limitation was measured, and the right-hand column is the work it asks for.

measured limitation	what closes it	cost
No evaluation with people	Run the study: 6 to 10 people	calendar
The deployed model is a lookup table	Inputs that vary with the headline	low
Coverage of 88.5%, and 50% in the worst case	Human judgement over a sample	calendar
Reads the headline, not the body	Read the body of the articles	new data
Drift: the input it depends on most is the one that drifted most	Re-train with what is observed	low

Two more: the delay is the *source*'s and not the infrastructure's (≈353 min to detect, 5 s to deliver); and **nothing is ever re-trained**, because the case base grows on its own and the weights do not.

The three questions, answered on screen



1. Is it unusual?

In words, before the number: 141 of the last 250 days moved this much or more. The verdict is *"Quiet"*, and it comes before any value.

2. Was it the company?

All of it was. The stock rises 0.99% while the market (−0.47%) and the sector (−0.54%) pulled the other way: the company's part is +2.00%. And the line below says this company's fit sits **below** the median.

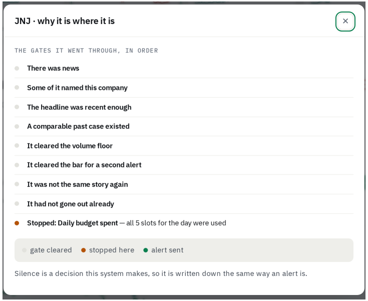
3. Has it happened before?

The days marked on the curve, and the precedents inside the text of the alert.

Demonstration

The system is live. **[recording]**

a real news item, from the moment it arrives to the alert that goes out



The path of one candidate through the gates, to where it stopped. **It is the part a live demonstration cannot show**: nine out of ten scans send nothing, and the silence is written down like an alert.

What I learned

1. **The baseline matters as much as the model.**
   The most useful result of the work came from looking at what I was comparing against.
2. **Where a model is placed matters as much as the model.**
   Mine worked on its own and stopped being useful behind two simple filters that were already doing the work.
3. **The simpler technique won three times.**
   Against Isolation Forest and against the model that reads the text, which are *simple against complex*. And a table of constants beat the deployed model, which is a different case: there the model was interpretable, and lost for **lack of information**.

The hard part was not choosing the most advanced technique.
It was building the evaluation **able to say that it was not needed**.

Thank you. Questions?